Design and Implementation of an Urban Mobility Database for Spatio-Temporal Traffic Analysis Using Advanced SQL

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Abstract

- Urban traffic generates a large amount of data from vehicles, road segments, traffic sensors, locations and time records. Managing this information efficiently is important for understanding congestion and traffic movement.

- We propose an urban mobility database that stores traffic information in a structured relational database, organized by roads, locations, vehicles, timestamps, speed and congestion conditions.

- Advanced SQL will be used to analyse the stored data - complex joins, subqueries, aggregation, analytical queries, indexes, views and materialized views - including time-based comparisons across locations and periods.

- The aim is to build a database system that can efficiently manage and analyse urban mobility data and provide meaningful traffic pattern reports.

Introduction

- Urban transportation systems continuously generate data — vehicle movement, traffic sensor readings, road locations, speed, volume, congestion levels and timestamps.
- When this information becomes large, simply storing it is not enough. A well-designed database is needed to:
  - Store the data efficiently
  - Maintain relationships between entities
  - Handle large numbers of records
  - Perform complex queries
  - Analyse traffic patterns
  - Produce useful reports
- Our project focuses on designing such a database using Advanced SQL.



Literature Review

S.No	Author's name	Title	Methods	Merits	Demerits
1	Wolfson & Xu (2010)	Spatio-temporal Databases in Urban Transportation	Survey of ITS data areas: routing, sensor data, GPS map-matching, dissemination	Broad foundation for why transportation needs spatio-temporal DBs	Survey only; no new schema or SQL implementation
2	Hajari & Hakimpour (2014)	A Spatial Data Model for Moving Object Databases	Spatial data model with UDTs + extended SQL for network-constrained moving objects (Oracle)	Concrete, DBMS-portable SQL extensions; combines spatial data, moving objects, SQL	Evaluated only on synthetic data, not real traffic data
3	Imawan et al. (2016)	Querying and Extracting Timeline Information from Road Traffic Sensor Data	Timeline data model + TQ-index for analytical queries over historical sensor data	Validated on real datasets (Busan ITS, Seattle freeway)	Tuned for congestion-timeline queries; less general for arbitrary spatial ops
4	Wolfson, Xu, Chamberlain & Jiang (1998)	Moving Objects Databases: Issues and Solutions	Identifies moving-object DB requirements (tracking, uncertainty, updates); proposes DOMINO framework	Foundational, highly-cited paper framing the moving-object DB subfield	Conceptual and dated; no experimental performance evaluation
5	Forlizzi, Güting, Nardelli & Schneider (2000)	A Data Model and Data Structures for Moving Objects Databases	Formal algebra of moving-point/moving-region types, pluggable into relational/OO DBMS	Rigorous, widely-cited reference model; maps cleanly onto SQL functions	High mathematical abstraction; heavy to fully implement for a course project

Literature Review

S.No	Author's name	Title	Methods	Merits	Demerits
6	Zhiming Ding (2008)	Moving Objects DBs Based on Dynamic Transportation Networks	Route-based model + indexing for moving objects on networks whose topology changes over time	Handles realistic road-network change; covers modeling, indexing, implementation	Dynamic-network complexity beyond what a semester project needs
7	Esheiba, Mokhtar & El-Sharkawi (2013)	Spatio-Temporal Queries for Moving Objects Data Warehousing	Spatio-temporal data-warehouse schema with OLAP-style measures over location history	Bridges moving-object DBs with OLAP aggregation; good model for GROUP BY-style analysis	Evaluated at small/conceptual scale, not on large real traffic datasets
8	Banaei-Kashani, Shahabi & Pan (2011)	Discovering Patterns in Traffic Sensor Data	Statistical spatio-temporal model mined over real high-resolution LA traffic sensor data	Real large-scale dataset; closely matches traffic pattern analysis	Focus on statistical pattern-mining, not DB/SQL mechanics
9	Ayabakan & Kilimci (2014)	Moving Object Databases: Indexing Algorithms	Comparative review of indexing structures: R-tree family, quad-trees, grid files	Good single-source overview to justify an index choice in schema	Survey/comparison paper, not a novel technique; short
10	Kwon, Lee & Lee (2002)	Indexing the Current Positions of Moving Objects Using the Lazy Update R-tree	Proposes LUR-tree: R-tree variant updating only leaf entries within existing bounding box	Classic, highly-cited indexing technique; concretely implementable (Oracle/PostGIS)	Optimized for small frequent moves; less effective if objects often leave their MBR

Work Flow of Proposed Model

01. Data Preparation -Traffic and mobility data is collected, cleaned, and standardized.

02. Database Design - A normalized relational database is designed with appropriate tables, keys, constraints, and relationships.

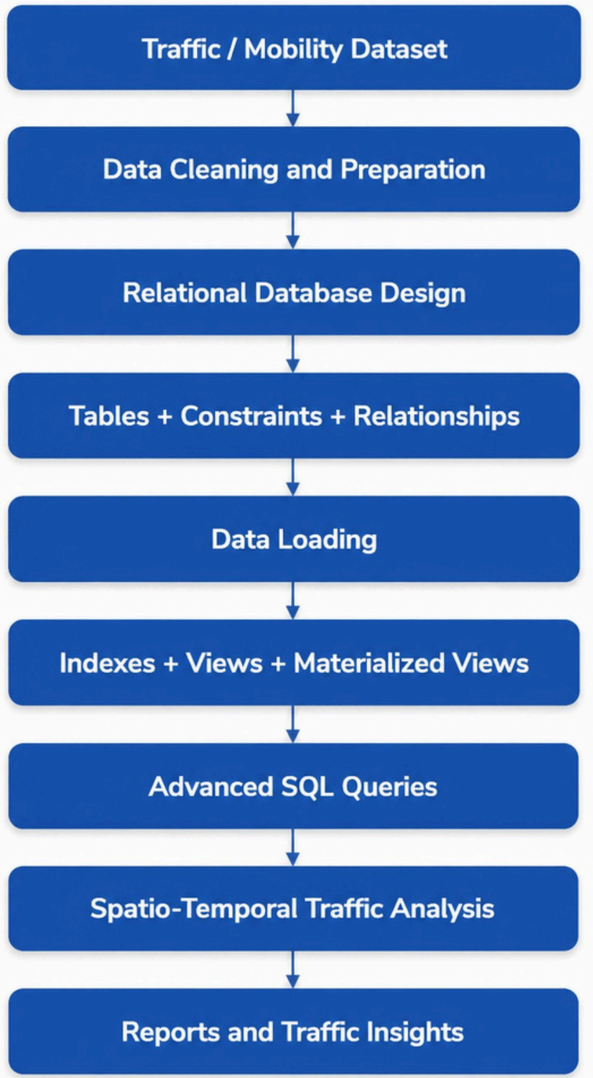
03. Data Integration - The prepared data is loaded and validated within the database.

04. SQL Optimization - Indexes, views, and materialized views are created for efficient data retrieval and reporting.

05. Advanced SQL Analysis - Joins, subqueries, CTEs, aggregate functions, and analytical queries are applied.

06. Spatio-Temporal Analysis - Traffic patterns are analysed across different locations and time periods.

07. Insight Generation - The results are transformed into traffic reports, trends, congestion patterns, and actionable insights.



Architecture Diagram

Urban Mobility Database for Spatio-Temporal Traffic Analysis Using Advanced SQL

Data Sources (Real World)

- **Traffic Sensors** (Vehicle count, Speed, Congestion)
- **Vehicles** (Movement, Trips)
- **Road Network** (Roads, Segments, Locations)
- **Time Information** (Date & Time)

Data Ingestion & Preprocessing Layer

- **Data Collection**
- **Data Cleaning** (Missing values, Errors)
- **Data Transformation** (Formatting, Standardization)
- **Data Loading** (Into Database)

Database Layer (Relational Database)

Urban Mobility Database

- ROAD
- VEHICLE
- ROAD_SEGMENT
- TRIP
- SENSOR
- TRAFFIC_OBSERVATION

(Structured and Related Tables)

Query & Analysis Layer (Advanced SQL)

- **SQL Queries** (Joins, Subqueries, CTEs)
- **Analytical Functions** (Ranking, Aggregation)
- **Views & Materialized Views** (Reusable Reports)
- **Indexes** (Query Optimization)
- **Spatio-Temporal Analysis** (Location + Time)

Output Layer (Results & Insights)

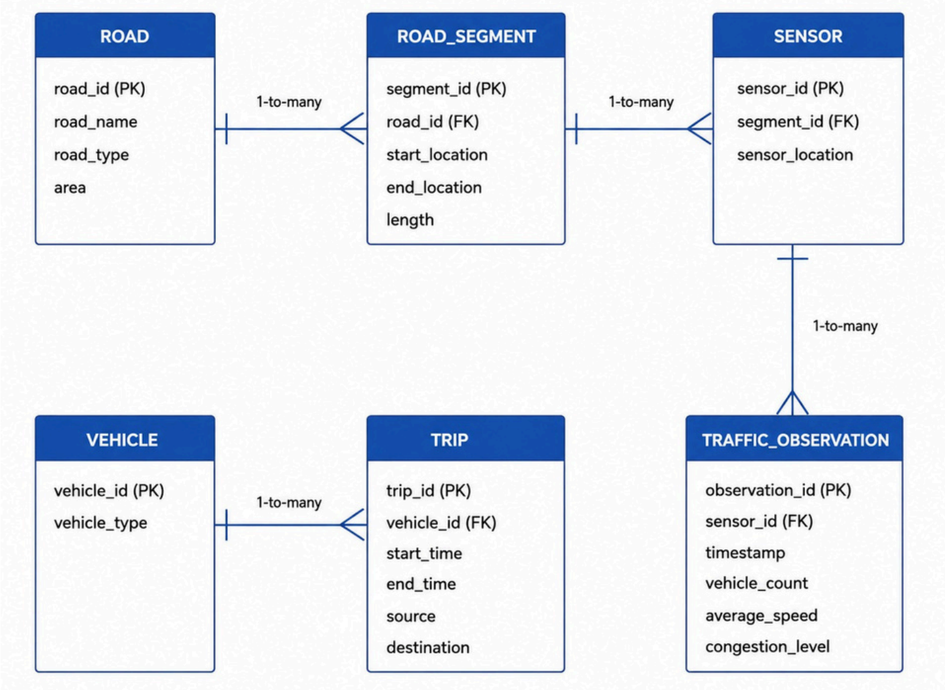
- **Traffic Analysis Reports** (Busiest Roads, Peak Hours)
- **Location-based Insights** (Spatial Analysis)
- **Time-based Insights** (Temporal Analysis)
- **Visualized Results** (Reports & Dashboards)

Users

- **City Planners** (Infrastructure Planning)
- **Researchers** (Traffic Pattern Analysis)
- **Administrators** (Database Management)

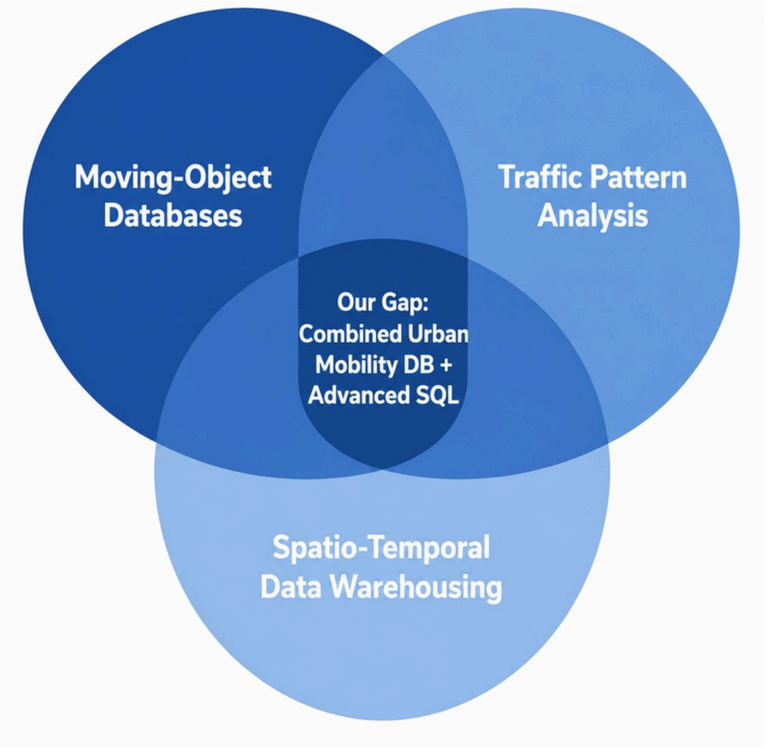
Proposed work -Explanation

- We propose an Urban Mobility Database containing related entities such as roads, road segments, vehicles, trips, traffic sensors, locations and traffic observations. Advanced SQL queries will then be used to analyse this information.
- **Main entities:**
  - ROAD (road_id, road_name, road_type, area)
  - ROAD_SEGMENT (segment_id, road_id, start_location, end_location, length)
  - VEHICLE (vehicle_id, vehicle_type)
  - SENSOR (sensor_id, segment_id, sensor_location)
  - TRAFFIC_OBSERVATION (observation_id, sensor_id, timestamp, vehicle_count, average_speed, congestion_level)
  - TRIP (trip_id, vehicle_id, start_time, end_time, source, destination)



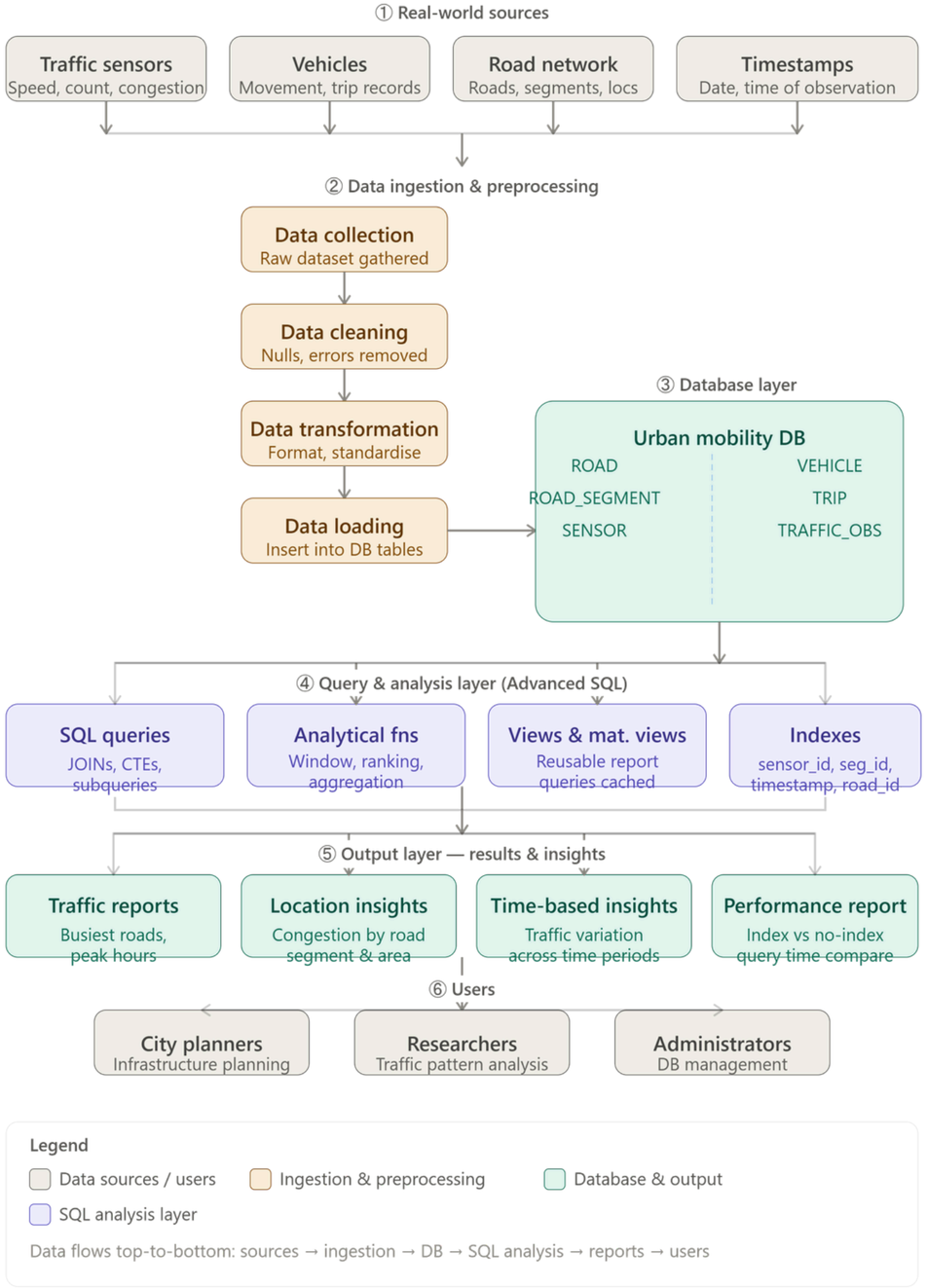
Research Gap

- Existing research provides strong solutions for individual parts of the problem — moving-object data models, specialized indexes, traffic pattern analysis, or spatio-temporal data warehousing — but not combined into one practical, student-level system demonstrated using Advanced SQL.
- **Our identified gap — a practical urban mobility database that combines:**
  - Relational database design
  - Traffic and mobility data with spatial and temporal attributes
  - Complex SQL analysis and query optimization using indexes
  - Views and materialized views
  - Historical traffic analysis



Data Flow

01 | Collect
Traffic and mobility data
02 | Prepare
Clean and standardize data
03 | Store
Organize data in relational tables
04 | Analyze
Apply Advanced SQL techniques
05 | Extract
Generate traffic patterns and insights
06 | Report
Present results for decision-making



References

1. Wolfson, O., Xu, B. (2010). Spatio-temporal Databases in Urban Transportation. Bulletin of the IEEE Computer Society Technical Committee on Data Engineering, 33(2), 18–25.
2. Hajari, H., Hakimpour, F. (2014). A Spatial Data Model for Moving Object Databases. International Journal of Database Management Systems, 6(1), 1–20. DOI: 10.5121/ijdms.2013.6101
3. Imawan, A., Indikawati, F.I., Kwon, J., Rao, P. (2016). Querying and Extracting Timeline Information from Road Traffic Sensor Data. Sensors, 16(9), 1340. DOI: 10.3390/s16091340
4. Wolfson, O., Xu, B., Chamberlain, S., Jiang, L. (1998). Moving Objects Databases: Issues and Solutions. Proc. 10th Int'l Conf. on Scientific and Statistical Database Management (SSDBM), pp. 111–122. DOI: 10.1109/SSDM.1998.688116
5. Forlizzi, L., Güting, R.H., Nardelli, E., Schneider, M. (2000). A Data Model and Data Structures for Moving Objects Databases. Proc. ACM SIGMOD, pp. 319–330. DOI: 10.1145/335191.335426
6. Ding, Z. (2008). Moving Objects Databases Based on Dynamic Transportation Networks: Modeling, Indexing, and Implementation. Springer LNCS, Vol. 4977, pp. 1–16.
7. Esheiba, L., Mokhtar, H.M.O., El-Sharkawi, M. (2013). Spatio-Temporal Queries for Moving Objects Data Warehousing. International Journal of Database Management Systems, 5(3), 1–13. DOI: 10.5121/ijdms.2013.5301
8. Banaei-Kashani, F., Shahabi, C., Pan, B. (2011). Discovering Patterns in Traffic Sensor Data. Proc. 2nd ACM SIGSPATIAL IWGS, pp. 10–16. DOI: 10.1145/2064959.2064963
9. Ayabakan, İ., Kilimci, P. (2014). Moving Object Databases – Indexing Algorithms. International Journal of Computer Theory and Engineering, 6(6), 455–459. DOI: 10.7763/IJCTE.2014.V6.909
10. Kwon, D., Lee, S., Lee, S. (2002). Indexing the Current Positions of Moving Objects Using the Lazy Update R-tree. Proc. 3rd Int'l Conf. on Mobile Data Management (MDM), pp. 113–120. IEEE Xplore.